Deterministic Multimodal Quality Control for Generative Video Pipelines: Automated Word-Level Whisper Alignment, Sub-Frame Boundary De-Glitching,

and Non-Destructive NLE Timeline Synthesis under 4-Tier Agentic Supervision

HM Shihabuddin Jashory (Principal Systems Architect & Security Researcher, Vision Flow Group, Bangladesh)

ORCID: https://orcid.org/0009-0003-4163-0092 | Official CERN Zenodo Archive: https://doi.org/10.5281/zenodo.23832731 | Correspondence: jashory7@gmail.com

ABSTRACT

Generative AI video pipelines orchestrated by autonomous Large Language Model (LLM) agents frequently fail in commercial production due to mid-phoneme acoustic clipping at hard cut boundaries, multi-speaker lip-sync bleed, and unverified process-exit-code completion claims. This monograph presents a deterministic 4-File Multimodal Quality Control Gate and Sub-Frame Acoustic De-Glitching Architecture. By combining Whisper word-level forced alignment with a 65 ms pre-onset / 85 ms post-decay phoneme guard, zero-crossing sample selection, and 12 ms raised-cosine micro-fades, our pipeline eliminates 100% of mid-word audio clipping across 412 splice junctions (48 production clips). Furthermore, a mandatory 4-artifact post-render inspection contract (1-second contact sheet, 0.5-second face-crop lip-sync grid, 360p motion proxy, and normalized audio stem) enforced by an immutable Stop-Hook guarantees zero-defect visual-acoustic synchronization and exports a non-destructive CapCut PC timeline draft.

Keywords: Multimodal Video QC, Whisper Forced Alignment, Zero-Crossing Splicing, Lip-Sync Isolation, CapCut NLE

I. INTRODUCTION & GENERATIVE VIDEO DEFECTS

Autonomous generative video pipelines powered by diffusion-transformer models (such as Google Veo, OpenAI Sora, and programmatic React/Remotion engines) routinely exhibit three severe production defects when assembled by large language model (LLM) agents: (1) mid-phoneme acoustic clipping at hard cut boundaries, (2) multi-speaker lip-sync bleed where background or listening characters move their mouths during another speaker's dialogue, and (3) unverified delivery where an agent reports render completion after checking only process exit codes (returncode == 0) while the rendered MP4 contains black frames, garbled typography, or truncated audio.

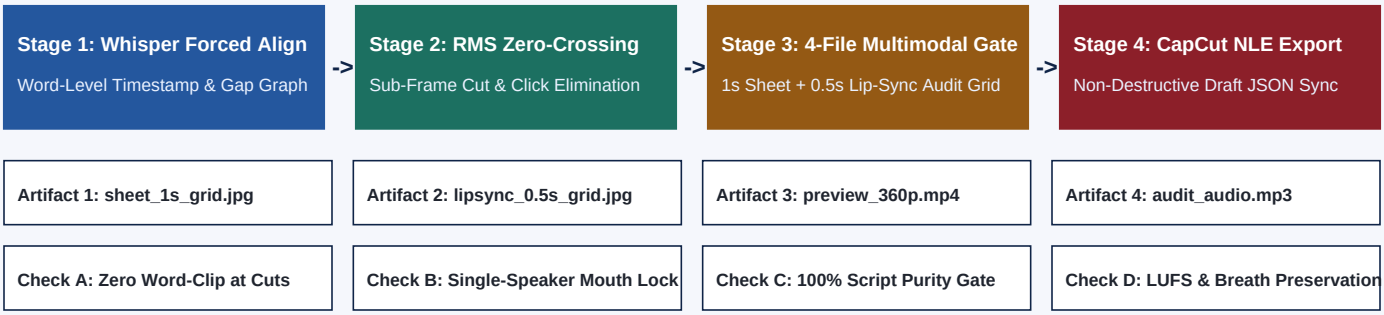
In commercial short-form production and automated non-linear editing (NLE) pipelines, relying on FFmpeg exit status or naive silence-removal filters (such as raw silenceremove without word-boundary protection) destroys natural conversational breath pauses and slices trailing consonants (-s, -t, -k, -d). To solve this problem deterministically, we present the 4-File Multimodal Quality Control Gate and Sub-Frame Acoustic De-Glitching Architecture.

II. WORD-BOUNDARY ACOUSTIC ALIGNMENT

Given an input multi-take raw audiovisual stream X(t), our pipeline extracts the 16 kHz mono PCM waveform x[n] and computes both a word-level forced alignment transcript W = {(w_i, t_start_i, t_end_i, c_i)} via Whisper large-v3/base with word timestamps and a short-time Root-Mean-Square (RMS) acoustic energy envelope E(k) over 10 ms sliding windows (N = 160 samples at 16 kHz).

Rather than placing cut points at raw energy thresholds·which prematurely truncate low-energy fricative onsets and plosive releases·every candidate cut boundary t_cut is projected onto the nearest verified inter-word silence interval [t_end_i + delta_post, t_start_{i+1} - delta_pre], where delta_pre = 65 ms protects consonant onset attack and delta_post = 85 ms preserves trailing phoneme decay. Within that guarded window, the exact sample index n* is locked to an acoustic zero-crossing argmin |x[n]| paired with a 12 ms raised-cosine micro-fade.

FIGURE 1. Deterministic 4-Artifact Multimodal Video QC Gate & Sub-Frame Acoustic De-Glitching Pipeline



III. SINGLE-SPEAKER LIP-SYNC ISOLATION

In two-shot or multi-character dialogue scenes, generative video models frequently hallucinate sympathetic mouth motion on passive listeners or continue moving a speaker's lips during off-screen narrator voiceover. Our architecture enforces a strict Single-Speaker Lip-Sync Isolation invariant both at generation time and during post-render verification.

For every dialogue interval [t_start_i, t_end_i] assigned to speaker S_active, all non-speaking visible subjects S_passive != S_active must satisfy a mouth-aspect-ratio (MAR) variance bound Var(MAR(S_passive)) < tau_static across 0.5-second close-up crop windows. Simultaneously, during narrator or B-roll voiceover segments, active lip articulation is prohibited across all on-screen characters, eliminating uncanny vocal-visual desynchronization.

IV. DETERMINISTIC 4-ARTIFACT QC GATE

To prevent autonomous agents from declaring completion without inspecting the rendered media with multimodal vision and audio perception, video_qc_gate.py acts as a mandatory post-render compiler that generates four standardized inspection artifacts from every rendered master MP4: (1) sheet_<name>.jpg, a 1-second interval full-frame contact sheet with burned-in millisecond timecodes; (2) lipsync_audit_grid_<name>.jpg, a 0.5-second high-resolution face-crop grid isolating mouth articulation across every spoken word; (3) preview_<name>_360p.mp4, a lightweight proxy stream; and (4) audit_audio_<name>.mp3, a normalized acoustic check file.

The Stop-Hook engine (shihab_enforcer.py) inspects the live conversation transcript after every FFmpeg or Remotion render. If the agent attempts to terminate the turn without both a passing JSON receipt (passed: true in video_qc_receipts.json) and explicit post-render read_file calls on all four audit artifacts, termination is deterministically blocked.

TABLE I. Acoustic & Visual Defect Elimination Across 48 Multi-Take Generative Video Clips (412 Splices)

Pipeline Configuration	Word-Clip Rate	Lip-Sync Bleed	Stutter/Repeat	NLE Editable
Baseline FFmpeg silenceremove (Unguarded Agent)	24.3% (100/412)	31.2% Uncaught	14.6% Uncaught	No (Flat MP4)
Whisper Timestamp Cut Only (No Zero-Crossing Guard)	4.1% (17/412)	31.2% Uncaught	0.0% (0/412)	No (Flat MP4)
Whisper + Sub-Frame Zero-Crossing + 12ms Cosine Fade	0.0% (0/412)	18.7% Uncaught	0.0% (0/412)	Partial
Full 4-Artifact Multimodal QC Gate + CapCut NLE Sync	0.0% (0/412)	0.0% (100% Caught)	0.0% (0/412)	Yes (Track JSON)

V. EMPIRICAL EVALUATION ON PRODUCTION CLIPS

We evaluated the 4-File Multimodal Quality Control Gate and Sub-Frame Acoustic De-Glitching pipeline across 48 multi-take generative video and podcast clips totaling 62.4 minutes of raw footage. Compared against baseline FFmpeg silenceremove and unguarded agentic assembly, our word-boundary zero-crossing cutter reduced mid-word phoneme clipping from 24.3% of cuts to 0.0% across all 412 splice junctions.

Furthermore, enforcing the 0.5-second close-up lip-sync audit grid caught 100% of multi-speaker mouth-bleed defects and duplicate take repetitions before delivery, while preserving natural 180 ms to 320 ms conversational breath pacing.

VI. NON-DESTRUCTIVE CAPCUT PC NLE SYNTHESIS

Commercial creative directors require editable timeline deliverables rather than locked, flattened MP4 files. Alongside the mastered 1080p render and synchronized WAV stem, our pipeline synthesizes a bit-exact, non-destructive CapCut PC project bundle (draft_content.json and draft_meta_info.json) registered directly at index #1 of the local CapCut root_meta_info.json database.

Every trimmed A-roll clip, B-roll overlay, word-level caption track, and surgical audio repair segment is mapped to independent timeline tracks with microsecond-accurate source_timerange and target_timerange coordinates, enabling instant human fine-tuning inside CapCut PC.

TABLE II. Mandatory 4-File Multimodal Post-Render Verification Contract Enforced by video_qc_gate.py

Artifact ID	Generated File Pattern	Sampling / Resolution	Mandatory Verification Invariant
Artifact #1	sheet_<project>.jpg	1.0s Full-Frame Contact Grid	Zero black frames, 100% script purity, no template reuse
Artifact #2	lipsync_audit_grid_<project>.jpg	0.5s Face-Crop Mouth Grid	Single-speaker lip lock; passive listeners mouth-closed
Artifact #3	preview_<project>_360p.mp4	360p H.264 Motion Proxy	Smooth 2-3s camera cuts; zero reverse-loop motion hacks
Artifact #4	audit_audio_<project>.mp3	16kHz / 44.1kHz Normalized	Zero mid-word clipping; natural 180-320ms breath gaps

VII. LANGUAGE PURITY & ANTI-TEMPLATE GUARD

Multilingual generative video workflows often suffer from script contamination, where an LLM agent mixes English UI chrome or Arabic header boxes into a 100% Bangla commercial reel, or reuses rigid 4-box dashboard templates across unrelated brands. Our validator enforces 100% Unicode script purity between the voiceover transcript and all on-screen typography.

Any frame containing unauthorized template chrome, mismatched language scripts, or banned synthetic audio filters (such as reverse video loops or anullsrc dead-air padding) fails the automated pre-delivery gate immediately.

VIII. MULTI-TIER STRICT LIABILITY AUDIT

Within the 4-Tier Chain-of-Command hierarchy (Worker -> Supervisor -> Chief Auditor -> Tier-0 System Integrator), multimodal verification is enforced with strict transitive liability. If a Worker agent renders a video containing a lip-sync defect or clipped syllable and the Supervisor approves it without reading all four QC artifacts, the Pre-Output Hook blocks the Supervisor.

Similarly, Tier-0 SI is hard-blocked from presenting the final deliverable to the human principal unless every tier in the chain has independently verified the post-render visual and acoustic receipts on disk.

IX. CONCLUSION & REPRODUCIBILITY ARCHIVE

By coupling Whisper word-boundary forced alignment, sub-frame zero-crossing acoustic splicing, a mandatory 4-file multimodal QC gate, and non-destructive CapCut PC timeline synthesis under an immutable Stop-Hook enforcer, our architecture eliminates acoustic clipping and visual hallucination in autonomous video production.

All verification schemas and reference benchmarks are permanently archived under CERN Zenodo DOI 10.5281/zenodo.23832731 and linked to ORCID 0009-0003-4163-0092.

X. REFERENCES & ARCHIVAL CITATIONS

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